



SpaceX Dragon

Dragon Capsule was recovered with research from the International Space Station. <http://dgit.in/SpaceXdragon>



Heart neural network

Kardia from AliveCor is neutral network that works with your mobile to protect your heart. <http://dgit.in/TheKardia>

BRINGING BACK THE WOOLY

How Manny from Ice Age could one day be walking amongst us

Neha C V & Ronak Gupta
feedback@dgit.in

When Jurassic Park released in 1993, moviegoers were spell-bound looking at the dinosaurs brought back to 'life on screen'. Micheal Crichton's science fiction book had a fanciful and thrilling plot that translated into pure cinematic joy. Scientists somehow revive T-Rex and Velociraptors and all hell breaks loose. Besides entertaining the hell out of cinema lovers, the movie planted everyone with an idea. Can we bring back extinct animals from the dead?

As it turns out, we probably can. Scientists at Harvard are trying their best to revive the Woolly Mammoth (Remember Manfred "Manny" from Ice Age). Woolly Mammoths are one of the most easily recognizable extinct creatures and were wiped out in one of the five mass extinction events. And now, we can possibly see them in flesh - thanks to a revolutionary tool that can modify the code of life - DNA. This tool, CRISPR, has been making the news lately for being a means of genetic engineering that precisely modifies available DNA samples with minimum errors, enabling scientists to model and find a cure for many genetic diseases. But how is this achieved? We have broken down the process of this seemingly magical technology in the November issue of Digit and although

the inner workings of this process seem sound enough, there is a nagging question that remains - where will the DNA of the "dead" animals come from?

The DNA Question

The Jurassic Park technique of mining dinosaur DNA from fossilised mosquitoes doesn't quite work in reality, as Dino DNA doesn't quite survive the vagaries of age. In the case of Mammoths, being from the Ice Age comes with its own benefits. The Woolly Mammoths that went extinct during this period got buried under huge chunks of ice and snow leading to proper preservation of their tissues. One such specimen, that of a 39,000-year-old female baby mammoth was discovered in Siberia in 2010. The specimen, named Yuka has been the source of all information we need to reconstruct these species. Equipped with the power and precision of CRISPR, the DNA encoding for mammoth features is being engineered into the embryo of an Asiatic elephant to produce a hybrid - a Mammophant.

The chosen one

The reason for choosing an Asiatic elephant over its cousin - The African Elephant, for making the hybrid is

that the Asiatic Elephant is genetically closer to the Mammoth, similar to how humans are more closely related to chimpanzees than to gorillas. This reduces the number of genetic changes needed. The group of scientists at Harvard have made 15 edits in the Asiatic elephant DNA and they hope to increase it to 45 edits soon to make the Mammophant as close to a mammoth as possible. With these edits in place, they then wish to grow this hybrid in an artificial womb in a laboratory. That is quite a massive task. The time taken



An artist's impression of the Woolly Mammoth

for a mammoth/elephant embryo to grow is around 660 days, and using current state of art only mice embryos have been cultured this way, that too for only 10 days - out of which half the time they originally spend within their mother's womb. Is there any reason why scientists want to embark on this difficult line of research - beyond the novelty factor?

(CC 2.5, credit: Mauricio Anton - Wikipedia)